

Software Development Fundamentals (SDF) – I

ODD 2023



Lecture 2

Jaypee Institute of Information Technology (JIIT), Noida

Outline of Lecture

- 1. Elevator Control System**
- 2. Puzzle Set 1 (Magic Square) – Design the Logic**
- 3. Puzzle Set 1 (Count the Lines) – Design the Logic**
- 4. Puzzle Set 1 (Count the Squares) – Design the Logic**
- 5. Exercises**

Elevator Control System



Source:

<https://www.insightssuccess.in/wp-content/uploads/2019/12/Elevating-Lifestyle.jpg>



Source: <https://i.ytimg.com/vi/rXJUFCbKWQM/maxresdefault.jpg>

Lift / Car (carry passengers)

- Moves up / down; Moves at different speed; Stops at floors
- Doors open/close at floors; emergency opening of doors

Elevator Request System (ERS)

Buttons for Elevator Request System (in Hall area)

1. Hall Call Buttons (HCB) – Two Types: Up and Down.
GF □ Up; TF □ Down; Remaining Floors □ Both HCBs
2. Indicators – At each floor. Two types (Direction/Position)

Elevator Control System (ECS)

Buttons for Elevator Control System (inside the Lift)

1. Lift Buttons – Floor Buttons (GF to TF); Door open/close buttons; Start/Stop buttons, *etc.*
2. Indicators – Two types (Direction/Position)
3. Emergency Buttons – Alarm, Calling buttons, *etc.*

Elevator Logic Controller (ELC)

1. Receives requests from ERS / ECS and serve the request
2. Apply Emergency Break whenever required

Elevator Control System

Flow of Events

ERS: Elevator Request System
ECS: Elevator Control System
ELC: Elevator Logic Controller

Elevator Request

- (1)** Passenger / User in Hall area **makes request** (REQ1)
 - presses Hall Call Button (Up or Down)
- (2)** **ERS transfers the request** (REQ1) **to ELC**
 - detailing with floor number
- (3)** **ELC adds the ERS request** (REQ1) **into the List of Floors to Visit** (LFV)
- (4)** Based on current LFV, **ELC decides**
 - **which direction** (up or down) **to move** to serve the next request (s)
 - move towards the requested floor (s)
 - to **stop** when the elevator is about to **reach to a floor in LFV** and **are along the route to REQ1**
- (5)** **ELC opens the elevator door** when it stops at a floor
- (6)** **ELC removes the served request from LFV**

Exercises

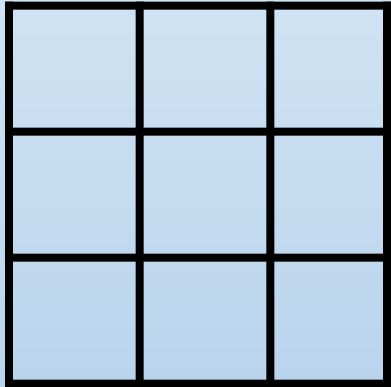
Q1. List down the flow of events when passenger inside the Lift selects the floor where he/she wants to move/stop (request is to be made by ECS)

Q2. List down the different scenarios or requirements in which ELC will apply the Emergency break

Will discuss your solutions/ approaches in Tutorial

Puzzle Set 1: Magic Square

Problem Statement: Given a $N \times N$ square grid (most popular is 3×3) and a set of nine continuous and distinct integers (often 1 to 9), it is desired to place the numbers into grids (one grid contains one number) so that summation of the numbers in each row, column, and diagonal (corner to corner) are same

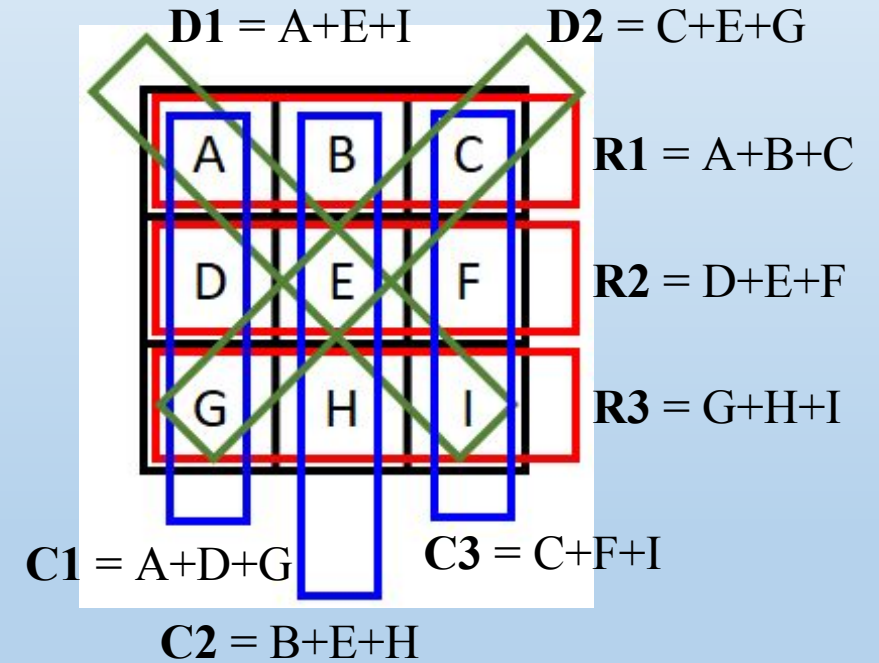


3×3 square grid

A	B	C
D	E	F
G	H	I

Place nine variables (A to I) in grids that hold distinct values, *e.g.* from 1 to 9

so that, $R1 = R2 = R3 = C1 = C2 = C3 = D1 = D2$



Puzzle Set 1: Magic Square

Possible solutions

for 3×3 square grid and integers from 1 to 9

8	1	6
3	5	7
4	9	2

4	9	2
3	5	7
8	1	6

2	9	4
7	5	3
6	1	8

6	1	8
7	5	3
2	9	4

SUM = 15

4	3	8
9	5	1
2	7	6

8	3	4
1	5	9
6	7	2

6	7	2
1	5	9
8	3	4

2	7	6
9	5	1
4	3	8

Let's DESIGN the LOGIC

Puzzle Set 1: Magic Square

DESIGN the LOGIC

A	B	C
D	E	F
G	H	I

Place nine variables (**A to I**) in grids that hold distinct values

Q1. What will be the Sum in each row or column or diagonal, if it is 3×3 square grid and numbers are from 1 to 9?

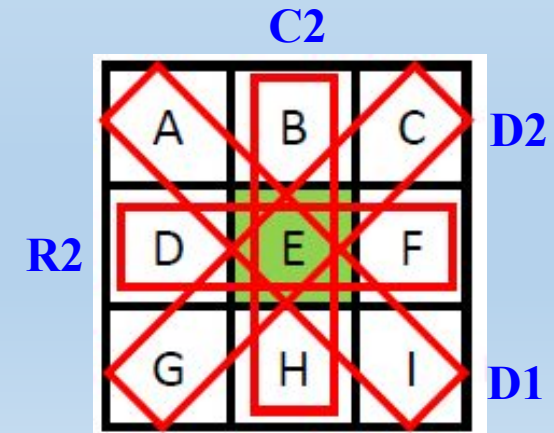
Ans. Sum will be 15

Argument. Sum of all numbers is, $\text{Sum(ALL)} = A+B+C+D+E+F+G+H+I$
i.e. for 1 to 9, it is 45 ($1+2+3+4+5+6+7+8+9$)
Sum(ALL) is equally distributed in 3 Rows
i.e. for 1 to 9, Sum in each Row will be 15 ($45 / 3$)
Hence, Sum in each Row/Column/Diagonal is 15 (for 1 to 9)

Q2. Out of 9 grids (if it is 3×3 square grid), a number from 1 to 9 should be entered into which grid first.

Ans. We should enter a number (from 1 to 9) into the Central Grid first.

Argument. Out of 8 sums (R1, R2, R3, C1, C2, C3, D1, and D2) number in central grid (see in Figure, it is E) appears in 4 sums (R2, C2, D1, & D2)



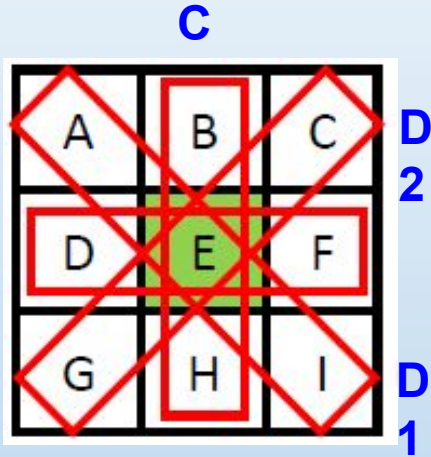
Puzzle Set 1: Magic Square

DESIGN the LOGIC

Q3. Which number (from 1 to 9) is to be entered into the Central Grid?

Ans. It should be 5

Argument. Number (E) in Central Grid appears in 4 Sums (R2, C2, D1, & D2)
Each Sum is 15, *i.e.* $R2 + C2 + D1 + D2$ is 60
 $(D+E+F) + (B+E+H) + (A+E+I) + (C+E+G) = 60$
 $(A+B+C+D+E+F+G+H+I) + (E+E+E) = 60$
i.e. 45 (from 1 to 9, sum of all numbers is 45) + 3 E = 60
Hence, E (number in Central Grid) is 5



Q4. Number at Central Grid is 5, place the remaining numbers in other grids

Ans. Let's explore

- The four sums, *e.g.* R2, C2, D1, & D2, collectively cover the numbers in each grid once, except the central grid
- R2 is 15, *i.e.* $D + E + F = 15$, OR, $D + 5 + F = 15$, Hence, pair of D & F can be (1, 9) or (2, 8), or (3, 7), or (4, 6)
- Similarly, pairs of (A & I) or (B & H) or (C & G) can be (1, 9) or (2, 8), or (3, 7), or (4, 6)

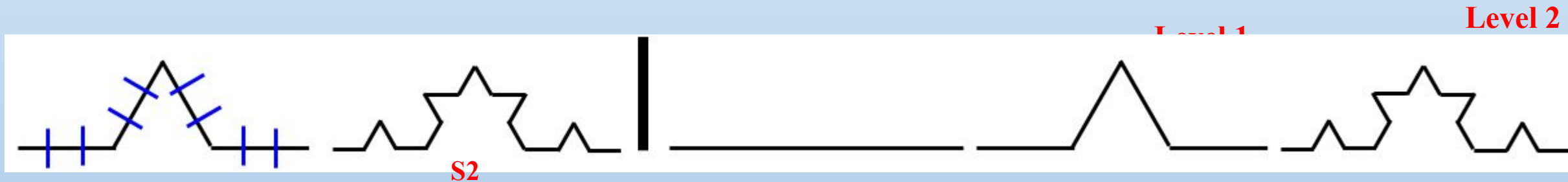
Design the remaining Logic by yourself to find out all possible solutions
Will discuss your Logic (with arguments) in Tutorial

Puzzle Set 1: Count the Lines

Problem Formulation: A line segment (LS) of length, L is divided into three equal parts. Considering the middle segment as base, an equilateral triangle is created as follows where base (which is the middle segment of LS) of the equilateral triangle is removed:



- As seen, a new shape (S1) is formed from LS having **FOUR** line segments, each of length $L/3$.
- Let us call, the transformation from LS to S1 as Level 1 transformation.
- In Level 2, each of the four line segments of S1 is further divided into similar manner and a new shape (S2) is formed from S1 having sixteen line segments, each of length $L/9$ as follows



Problem Statement: You have been given the initial shape as an equilateral triangle and the transformation rule as discussed above. Starting with initial shape (Level 0), repeatedly apply the transformation rules up-to Level 3. Find out the shape and the count of line segments at Level 3.

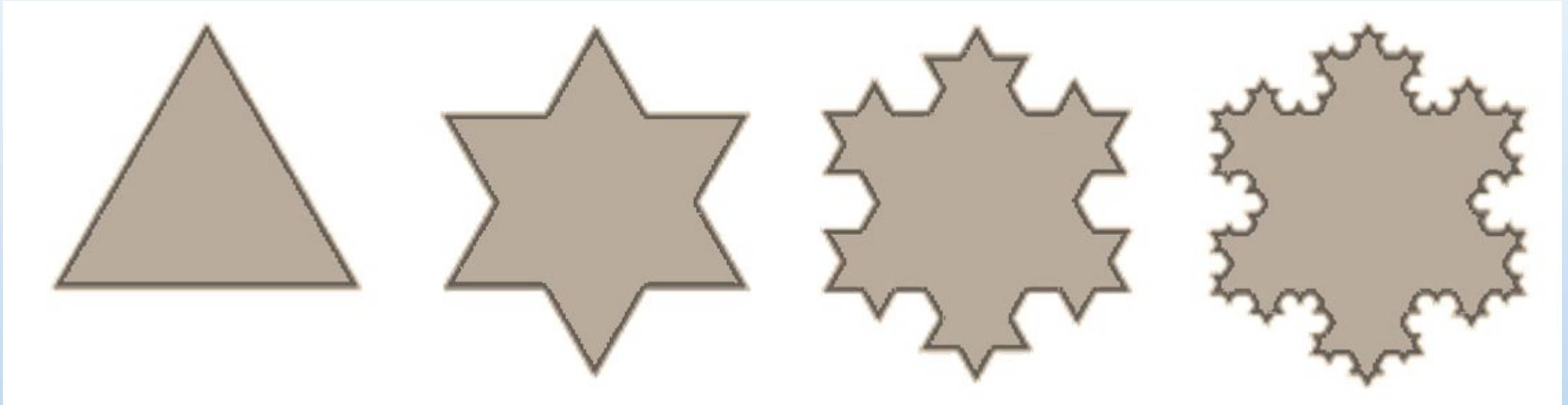


Level 0: Three Lines

Level 3: Shape = ?
Line Segment Count = ?

Puzzle Set 1: Count the Lines

Draw the Shape & Count the Lines



Level
0
Lines

Level
12
Lines

Level
48
Lines

Level
192
Lines

DESIGN the Logic: Each line segment is divided into 4 line segment in next level. Initial shape (Level 0) contains 3 lines.
Therefore, total lines at any level X (where X = 0, 1, 2,) will be

$$3 \times 4^X$$

$$\text{Level 0} = 3 \times 4^0 = 3 \text{ Lines}$$

$$\text{Level 1} = 3 \times 4^1 = 12 \text{ Lines}$$

$$\text{Level 2} = 3 \times 4^2 = 48 \text{ Lines}$$

$$\text{Level 3} = 3 \times 4^3 = 192 \text{ Lines}$$

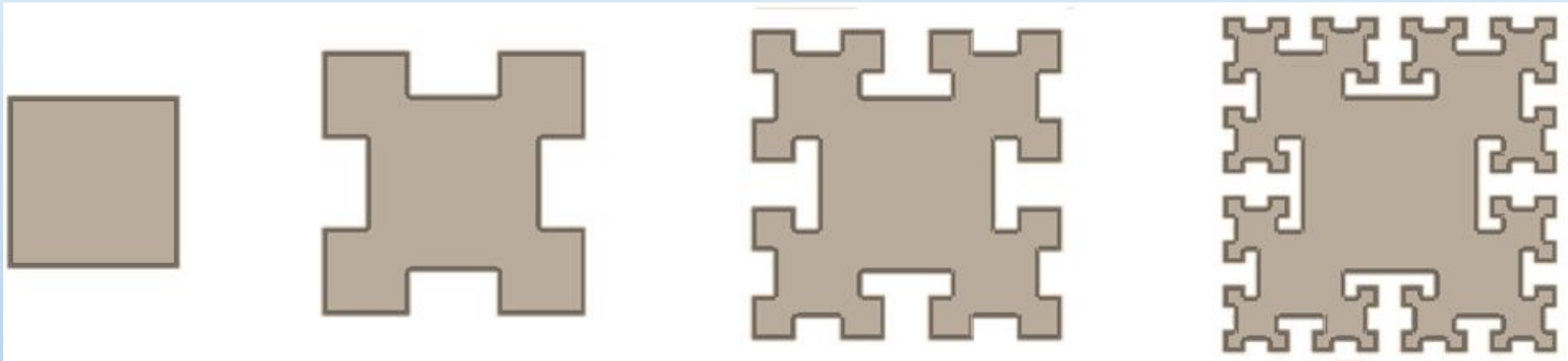
Puzzle Set 1: Count the Squares

Problem Statement: With initially given shape as Square, it is desired to repeatedly transform the shape in next stage/level as follows:

In next stage/level, **place another square centered at each convex corner** of the squares in the previous stage/level.

Side length of the newly placed square **should be half the side length** of the squares in the previous stage/level.

Starting with single square and mentioned transformation rules, how many new squares will be there at Level 3?



Level 0
1 Square

Level 1
4 New Squares

Level 2
? New Squares

Level 3
? New Squares

DESIGN the Logic: Each square in previous Level has 3 convex corner, where new squares to be placed in next Level.

Initially, there are 4 convex corners, therefore, new squares at any level X (where $X = 0, 1, 2, \dots$) will be

$$4 \times 3^{X-1}$$

$$\text{Level 1} = 4 \times 3^0 = 4 \text{ New Squares}$$

$$\text{Level 2} = 4 \times 3^1 = 12 \text{ New Squares}$$

$$\text{Level 3} = 4 \times 3^2 = 36 \text{ New Squares}$$

Exercises

- Q1.** Find out the count of possible solutions in Magic Square puzzle, if 5×5 square grid is given and numbers are from 1 to 25. Also compute the Sum in each row, column, and corner to corner diagonal.
- Q2.** Compute the Sum in each row, column, and corner to corner diagonal in Magic Square puzzle, if 3×3 square grid is given and numbers are from 7 to 15. Also find out all the possible solutions.
- Q3.** Can we create a Magic Square with user inputted random numbers? Say the user inputted nine numbers for 3×3 square grid are 1, 5, 6, 9, 12, 13, 17, 18, and 21.
- Q4.** In Count the Line puzzle, if length of one side of the equilateral triangle at Level 0 is 27 units, then what will be the perimeter of the transformed shape at Level 3?